University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Midterm Examination, Autumn 2025
Course Title: Digital Logic Design
Course Code: CSE0611215/CSE215

Marks: 20 **Time: 1*(one)* hour**

(Answer all questions)

Marks

1. **a)** A digital computer uses 8-bit 1's complement arithmetic for signed numbers. During a memory check, two values are read from different devices: **[04]**

- Device A reports the value X= D5 (hexadecimal).
- Device B reports the value Y= 73 (octal).

The system needs to compare the two device readings by performing X – Y and Y – X using 1's complement.

i. **Convert** both device values into their binary (8-bit) form.

ii. **Perform** the subtraction X−Y and Y−X using the 1's complement method and interpret the results in binary form.

b) An alarm security system function is given by **[04]**

$f(x,y,z)=x'y'z+x'yz+xy'$

Derive the reduced form of the function. Find the complement of the simplified function and also draw the circuit diagram.

c) **Differentiate** between standard form and canonical form of boolean algebra in a digital logic system. **[02]**

2. **a)** A university has an automatic gate system that opens based on three conditions: **[04]**

- A = The student has a valid ID card.
- B = The student has completed registration for this semester.
- C = The student is entering during office hours.

The gate will open if at least two of the conditions are satisfied.

Construct the truth table for the system and derive the Boolean expression using SOP and POS (Product of Sums) form.

b) A digital system has four input variables: **[06]**

- w = Sensor 1 (active if it detects motion)
- x = Sensor 2 (active if it detects heat)
- y = Time of day (0 = day, 1 = night)
- z = Security mode (0 = off, 1 = on)

The alarm will trigger (output = 1) for the following minterms:

$$F(w, x, y, z) \;=\; \sum (1, 3, 7, 11, 15)$$

Additionally, the system has some input combinations that will never occur (don't-care conditions):

$$d(w, x, y, z) \;=\; \sum (0, 2, 5, 8, 10)$$

Design the 4-variable K-map including the don't-care conditions. Express the simplified expression in both the Sum of minterms and the product of maxterms.